NESTED LOOP CALENDAR GENERATOR

Python Project Report

1.Overview

The Nested Loop Calendar Generator is a Python-based project designed to generate a calendar for a given month and year. It uses nested loops to display dates in a structured format, including days of the week. This project demonstrates the use of loops, conditional statements, and date calculations in Python.

2.Example Record Format

Input Field	Example Value
Year	2026
Month	10
Month Name	October
Number of Days	31
First Day	Thursday

3.Problem Statement

Creating a monthly calendar manually can be time-consuming. The Nested Loop Calendar Generator aims to generate a calendar automatically for a given month and year using Python nested loops. It displays the dates in a clear, organized format.

4.Objectives

1. To generate a calendar for a given month and year.
2. To display dates under the correct days of the week.
3. To implement nested loops in Python.
4. To identify the number of days in a month.
5. To improve programming and logical thinking skills.

5.Assumptions

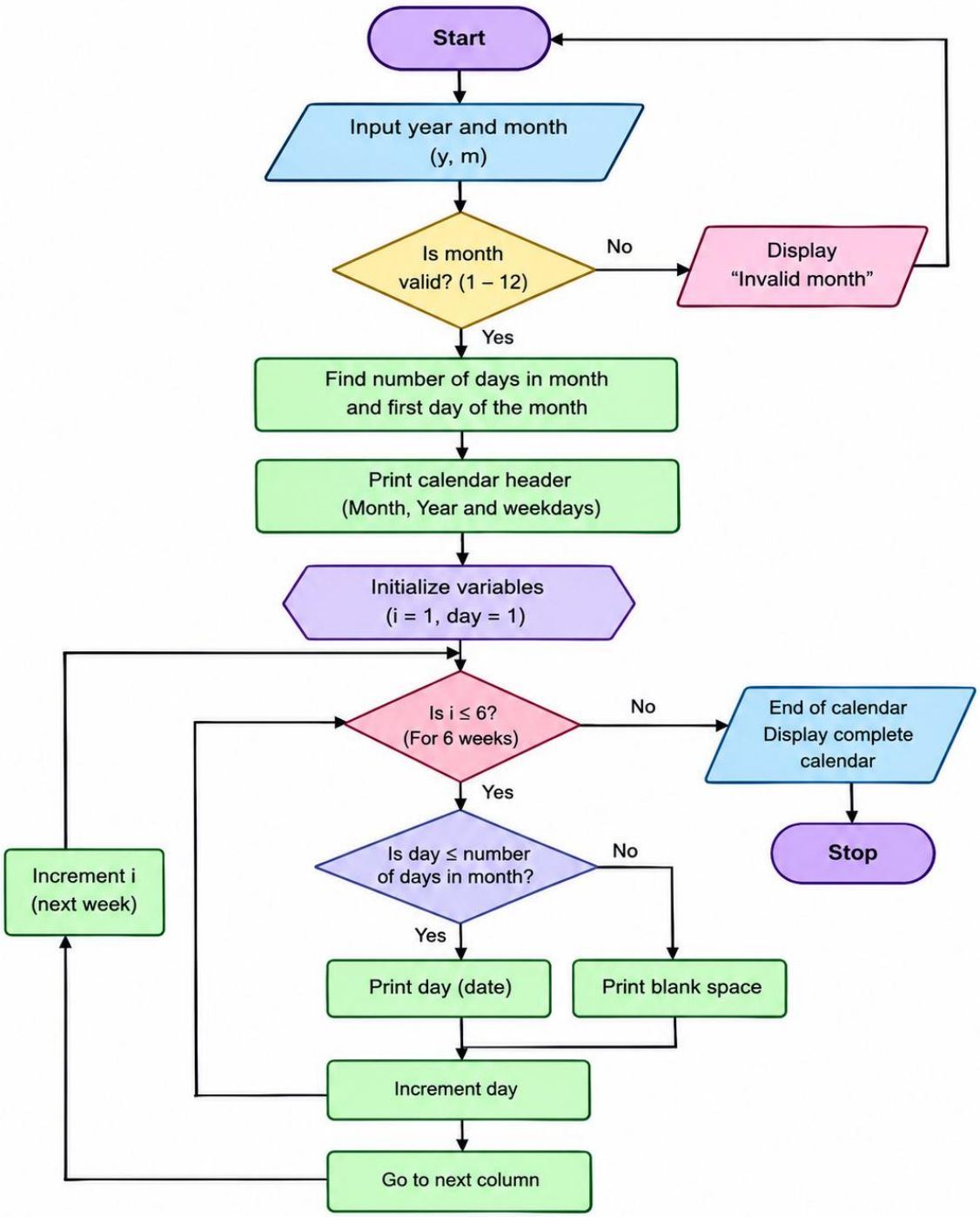
1. The user enters a valid year and month.
2. The month must be between 1 and 12.
3. The calendar follows the Gregorian calendar.
4. Leap years are handled correctly.

5. The calendar week starts on Sunday.

6.Algorithm

1. Start the program.
2. Input the year and month.
3. Validate the entered month and year.
4. Determine the number of days in the month.
5. Find the starting day of the month.
6. Use nested loops to arrange the dates in rows and columns.
7. Display the calendar with weekdays.
8. Stop the program.

7.Flowchart



8.Pseudocode

BEGIN

INPUT year and month

IF month is invalid

DISPLAY "Invalid Month"

```
  ELSE
    FIND number of days and first weekday
    DISPLAY calendar header

    FOR each week
      FOR each day of the week
        DISPLAY date or blank space
      END FOR
      MOVE to next line
    END FOR

    DISPLAY calendar
  END IF

END
```

9.Source Code

```
# NESTED LOOP CALENDAR GENERATOR

import calendar

def generate_calendar():
  print("=" * 35)
  print(" NESTED LOOP CALENDAR GENERATOR")
  print("=" * 35)

  try:
    year = int(input("Enter Year: "))
```

```
month = int(input("Enter Month (1-12): "))

if year < 1 or month < 1 or month > 12:
    print("Invalid year or month!")
    return

# Get calendar details
first_day, total_days = calendar.monthrange(year, month)

# Display month and year
print("\n", calendar.month_name[month], year)
print("Su Mo Tu We Th Fr Sa")

# Convert Monday-based weekday to Sunday-based
start_day = (first_day + 1) % 7

day = 1

# Outer loop for weeks
for week in range(6):

    # Inner loop for days of the week
    for weekday in range(7):

        if week == 0 and weekday < start_day:
            print("   ", end="")

        elif day <= total_days:
```

```
            print("{:2d} ".format(day), end="")
            day += 1

        else:
            print("   ", end="")

    print()

    if day > total_days:
        break

print("\nCalendar generated successfully!")

except ValueError:
    print("Invalid input! Please enter numbers only.")


if __name__ == "__main__":
    generate_calendar()
```

10.Test Cases

Test Case ID	Input	Expected Output
TC01	Year: 2026, Month: 10	Displays October 2026 calendar
TC02	Year: 2024, Month: 2	Displays 29 days (leap year)
TC03	Year: 2025, Month: 2	Displays 28 days
TC04	Year: 2026, Month: 13	Displays invalid month message
TC05	Year: 2026, Month: 0	Displays invalid month message
TC06	Year: abc, Month: 5	Displays invalid input message

TC07	Year: 2026, Month: 4	Displays April with 30 days

TEST RESULT

The test cases verify calendar generation, leap year handling, month validation, and invalid input handling.

11.Output Screenshots

```
==================================
 NESTED LOOP CALENDAR GENERATOR
==================================
Enter Year: 2026
Enter Month (1-12): 10

 October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

Calendar generated successfully!
```

12.Conclusion

The Nested Loop Calendar Generator is a Python-based project that generates a calendar for a given month and year. It displays dates in the correct order using nested loops and conditional statements. The project also handles leap years and invalid inputs. It demonstrates the practical use of Python programming concepts, including loops, functions, and input validation. This application makes calendar generation simple, accurate, and efficient.